

Fisetin regulates obesity by targeting mTORC1 signaling.

Fisetin, a flavonol present in vegetables and fruits, possesses antioxidative and anti-inflammatory properties. In this study, we have demonstrated that fisetin prevents diet-induced obesity through regulation of the signaling of mammalian target of rapamycin complex 1 (mTORC1), a central mediator of cellular growth, cellular proliferation and lipid biosynthesis. To evaluate whether fisetin regulates mTORC1 signaling, we investigated the phosphorylation and kinase activity of the 70-kDa ribosomal protein S6 kinase 1 (S6K1) and mTORC1 in 3T3-L1 preadipocytes. Fisetin treatment of preadipocytes reduced the phosphorylation of S6K1 and mTORC1 in a time- and concentration-dependent manner. To further our understanding of how fisetin negatively regulates mTORC1 signaling, we analyzed the phosphorylation of S6K1, mTOR and Akt in fisetin-treated TSC2-knockdown cells. The results suggested that fisetin treatment inhibits mTORC1 activity in an Akt-dependent manner. Recent studies have shown that adipocyte differentiation is dependent on mTORC1 activity. Fisetin treatment inhibited adipocyte differentiation, consistent with the negative effect of fisetin on mTOR. The inhibitory effect of fisetin on adipogenesis is dependent of mTOR activity, suggesting that fisetin inhibits adipogenesis and the accumulation of intracellular triglycerides during adipocyte differentiation by targeting mTORC1 signaling. Fisetin supplementation in mice fed a high-fat diet (HFD) significantly attenuated HFD-induced increases in body weight and white adipose tissue. We also observed that fisetin efficiently suppressed the phosphorylation of Akt, S6K1 and mTORC1 in adipose tissue. Collectively, these results suggest that inhibition of mTORC1 signaling by fisetin prevents adipocyte differentiation of 3T3-L1 preadipocytes and obesity in HFD-fed mice. Therefore, fisetin may be a useful phytochemical agent for attenuating diet-induced obesity. Copyright © 2013 Elsevier Inc. All rights reserved.